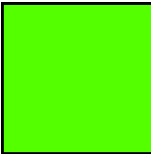
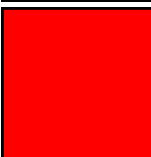
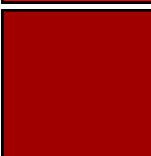
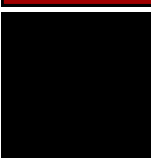


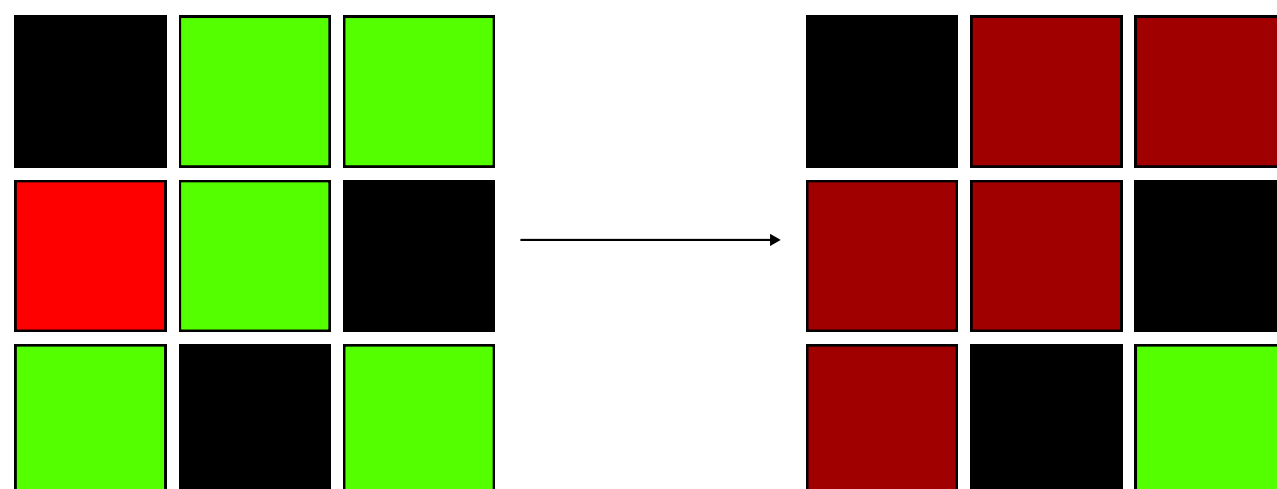
Can modelling social networks as physical systems help us find actionable insights?

Nishit Singh
BITS Pilani
February 2026

Percolation Models and communities

It is easier to see percolation theory in action than it is to describe it. Imagine a forest with a set density of trees, the first row of which are set to fire. A simple set of rules :

	Alive tree - burn if immediate neighbour is burning
	Burning tree - convert to burnt tree
	Burnt tree - does not interact
	Empty spot - does not interact

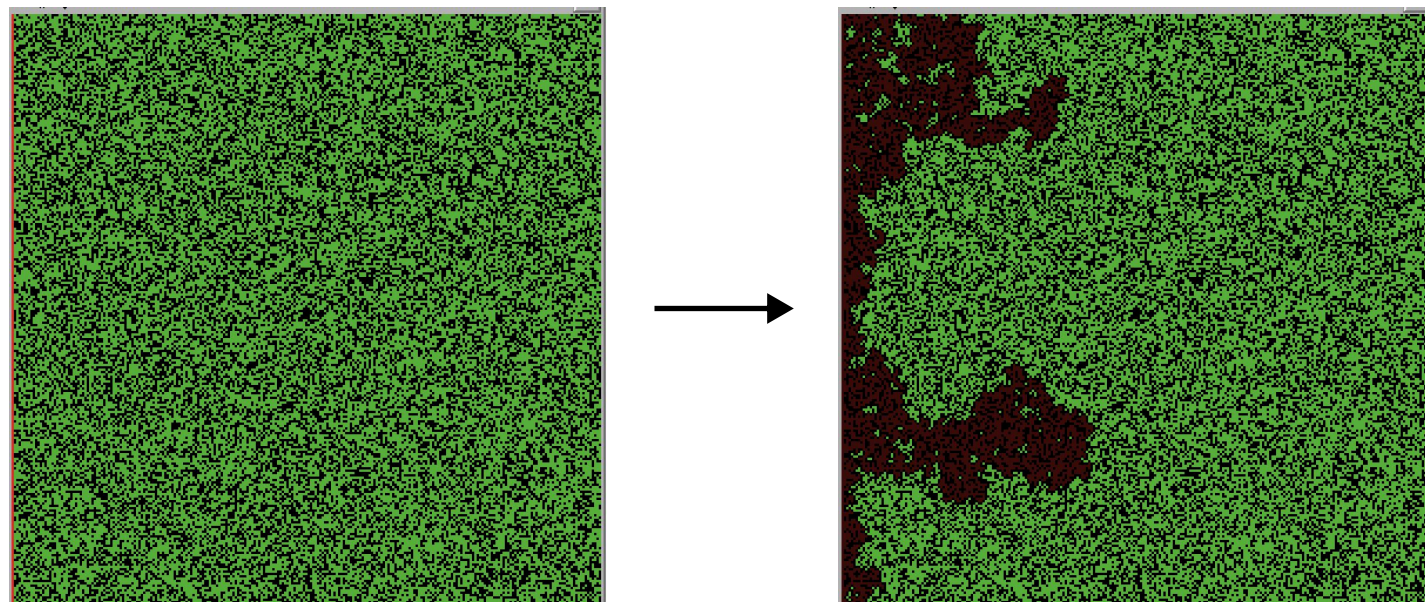


Initial state

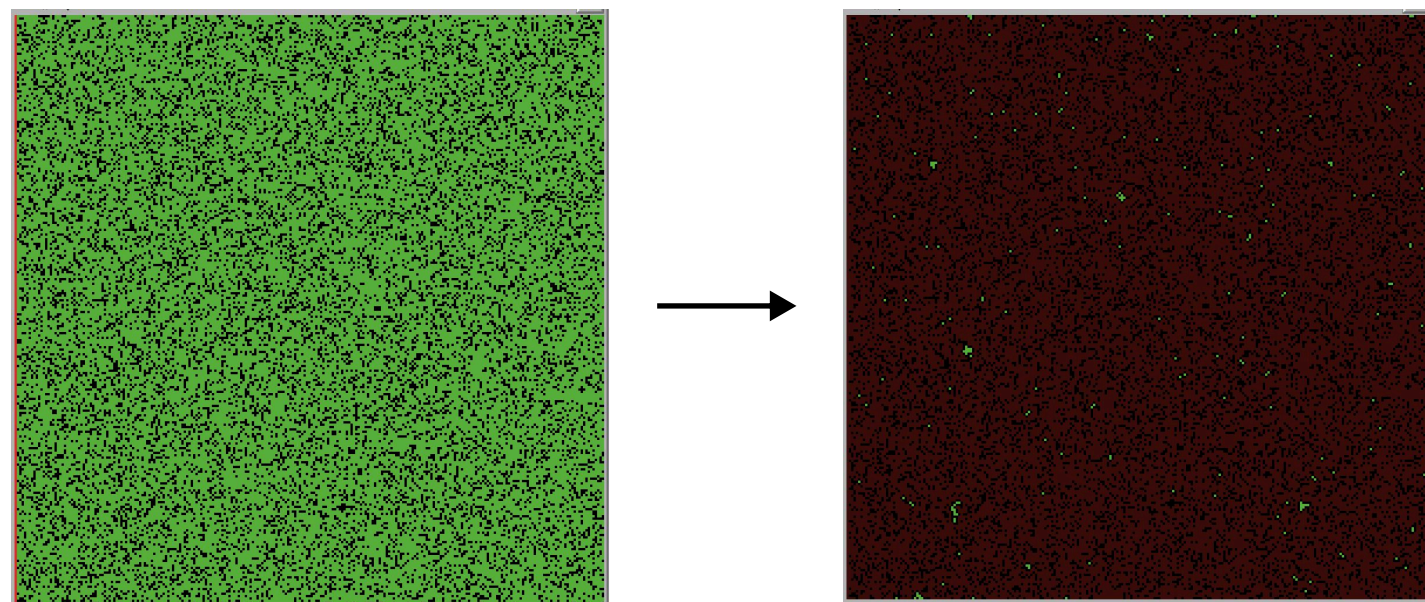
Final state

Percolation Models and communities

When coded up and run on bigger initial states, this program yields interesting results and even more interesting questions. ([Here](#) are videos of simulations running)



Forests less dense are more robust to forest fires, trivially.



Could the “architecture” of these forests tell us something about fragmentation in communities?

Percolation Models and communities

How does this connect to social networks?

The forest experiment shows us two things - (1) There exists a density after which, the fire will propagate from one end to the other, no matter how the trees are planted, and (2) even when the fire propagates fully, some clusters emerge in the final stage unharmed (they were unreachable from any of the initial positions).

Connecting to social networks, if we were to take Twitter as an example, there exists a threshold amount of mutual connections after which an “opinion” will be sure to flow from one point to another (I am not saying anything about the user adopting the opinion), and there is a way to moderate/promote accounts strategically to conserve subspaces so that they are unharmed by the flow of such opinions.

Percolation Models and communities

Percolation theory studies the emergence of connectivity in random systems. As a paradigm for random and semi-random connectivity, percolation models play a key role in network science, with concepts like the emergence of giant clusters, finite-size scaling, and mean-field methods employed to quantify robustness, epidemic spreading, and community dynamics.

A toy analogue to study social networks through the lens of percolation theory -

Nodes	Individual users
Edges	Social connections
Occupation probability	Probability that a node is healthy (p) or unhealthy ($1-p$)
Percolation threshold	Critical fraction of healthy / unhealthy nodes before fragmentation into smaller echochambers

Percolation Models and communities

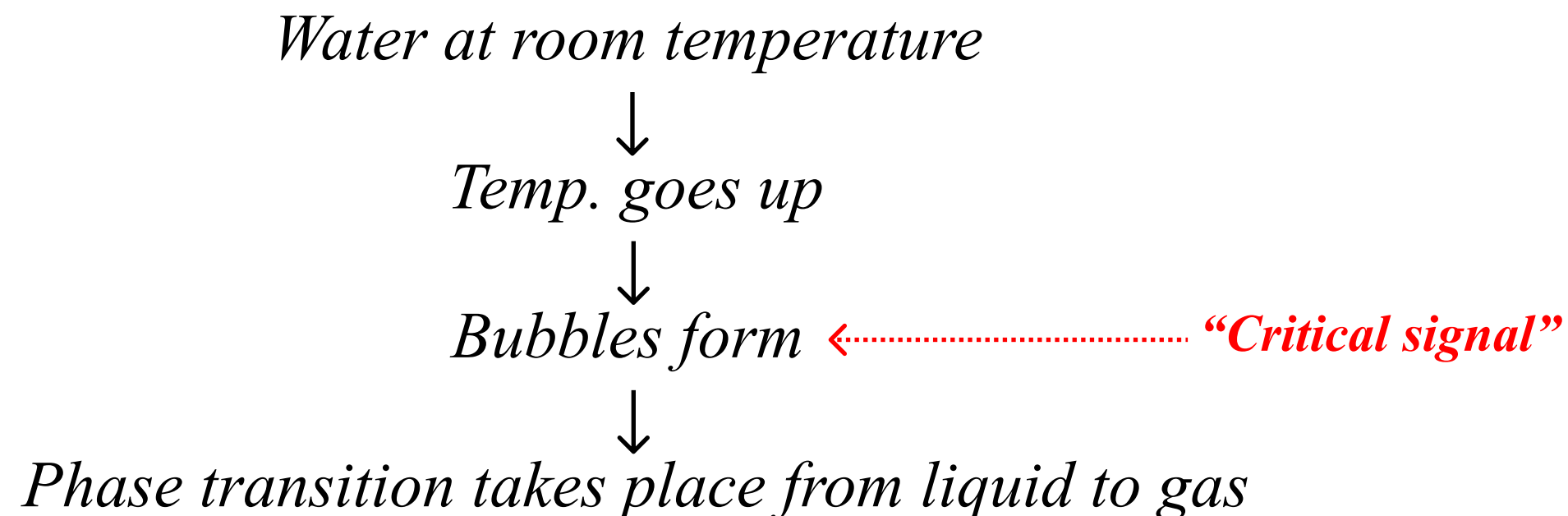
Finally, the research proposition.

Is there a way to model social networks as a graph state with weighted edges and using the rich landscape of percolation theory to get insights about the “architecture of online spaces (who has access to what)”, “moderation (which users/forums when moderated will have the biggest effect on the overall climate)”, and the impact of affordances?

Furthermore, is there an opening to utilise AI methods to retrieve these inferences quicker? Is there a way to train a model to take as input a graph state with transient edge weights (people identifying more with a certain section/user) and get insights as to when the network will break into smaller, disconnected graphs (echo-chambers and unhealthy discussion forums)?

The Ising Model → Social Networks

The canonical example for a system undergoing phase transitions is boiling water.

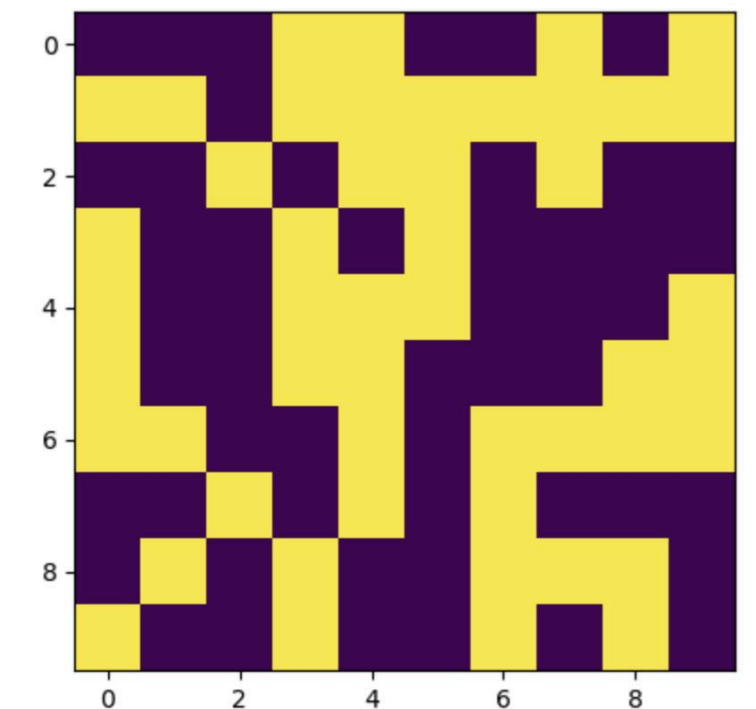


Is there a way to map the science we know to social networks in order to identify these “critical signals” systems exhibit before reaching a tipping point?

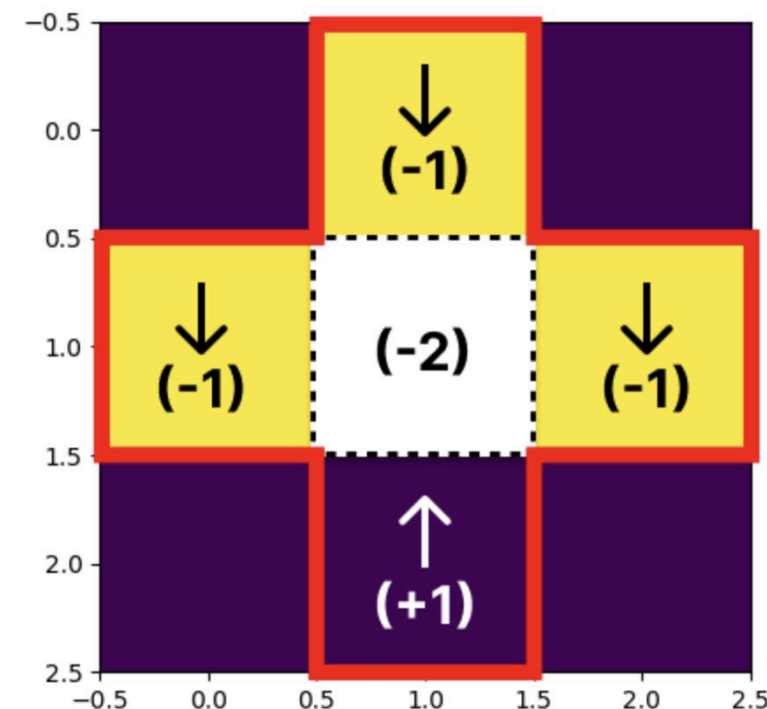
The Ising model is a method to simulate these tipping points in systems which interact with the properties of its surroundings and the relationships of its composite individuals with each other.

The Ising Model → Social Networks

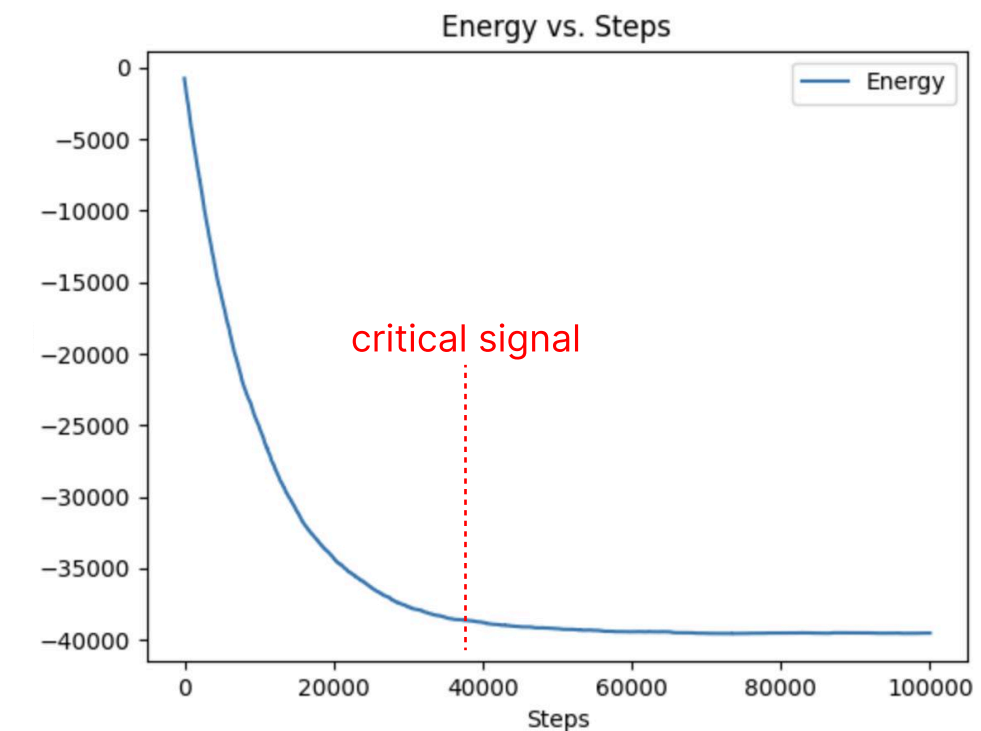
Here is the working of a simplified Ising Model (taken from [my GitHub](#)).



Matrix representation of a 10x10 system with 50% up-spins



Energy computation of a 3x3 state with 50% up-spins



Generate a random state

Each state has an “energy”

Purple → +1

Yellow → -1

Find the energy of
each particle

The new energy of each
particle is defined as the sum
of the energies of its
neighbours

Minimise the energy
of the system

“Flip” a particle to go from
(-1) → (+1) or (+1) → (-1)
and then undo the flip if the total energy rises,
or keep the flip with probability
 $p = p(\Delta E, T)^*$

*The exact function of P is a result from statistical mechanics

The Ising Model → Social Networks

How does this map to social networks?

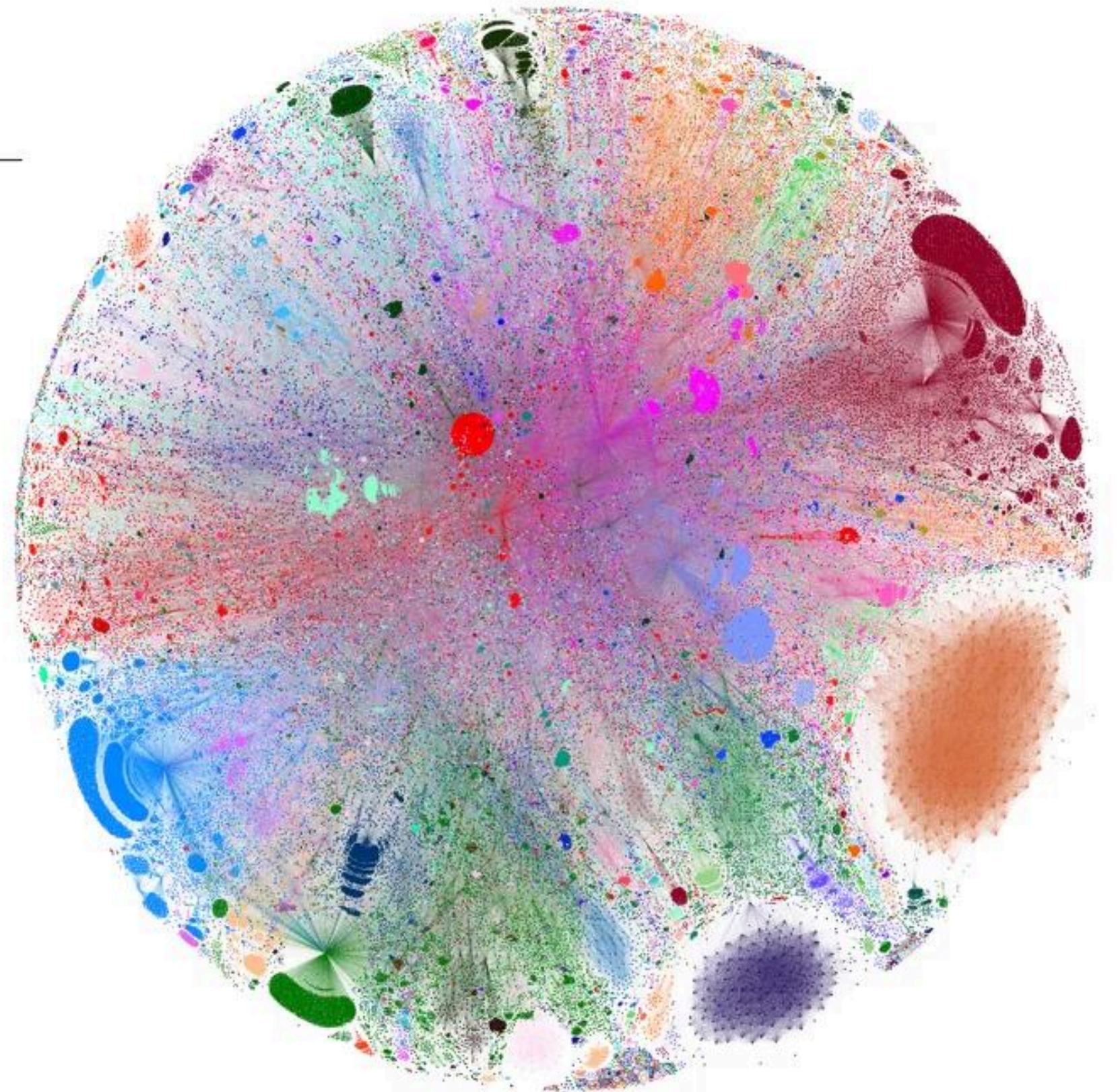
The Ising Model tells the story of a neighbourhood where each particle is influenced by the particles in a certain radius. Mapping social networks as a neighbourhood where each node affects another and figuring out the analogues between social indices and physical variables leads to fruitful discovery. A hand-wavy example of a social index → physical variable mapping is given below.

Energy states ($\uparrow/\downarrow$)	binary political opinions (liberal/conservative, pro/anti)
Coupling strength (J)	social influence between connected individuals
External field (h)	media bias, algorithmic curation, or elite messaging
Temperature (T)	social noise, uncertainty, or openness to opinion change

The Ising Model → Social Networks

An incredible map of reddit from u/wojtek-graj, where each node is a subreddit, and an edge is a user part of both communities.

% of subreddits	Description
10.99	NSFW
7.81	Europe
5.88	USA, outdoors
5.09	Food, travel, cars, finance
4.76	Wholesome funny
3.43	Gaming
3.37	General popular content
3.11	General popular content
2.96	Music
2.70	STEM
2.45	General popular content
2.36	French
2.23	TV, Movies
1.70	Latin America, Africa
1.61	Anime
1.54	Conversation, advice
1.47	Programming, videography
1.40	SciFi, literature
1.29	Tabletop / card gaming
1.20	Cats
1.16	History, religion, collecting
1.13	Political left
1.10	Hardware
1.09	Germany
1.04	Fitness
1.03	Drugs



The Ising Model $\rightarrow$ Social Networks

Finally, the research proposition.

Can we model online communities like Ising Models, and use this simulation to detect early signs of cascading effects like the formation of extremist echo-chambers ([r/The_Donald](#))? Is having a custom physics-informed loss function with the Ising Model as a structural prior helpful in recognising these critical transitions?

Closing comments and references

I understand that a physics flavour to an already interdisciplinary laboratory is a big leap, but I hope I was able to get across my fascinations and curiosities to some extent.

The diagrams in the presentation are all original work from my past experiences, except the Reddit Observable Universe*. Happy to discuss the nitty-gritty!

I am also happy to give this presentation orally, and so I have kept the information density to be somewhere in between a formal research statement and a completely oral presentation.

*Coined in the comments section of the Reddit post about the Map of Reddit.

Baldassarri et al.

Ising model on clustered networks: A model for opinion dynamics

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⁵Vrije Universiteit Amsterdam, Amsterdam, The Netherlands

December 26, 2022

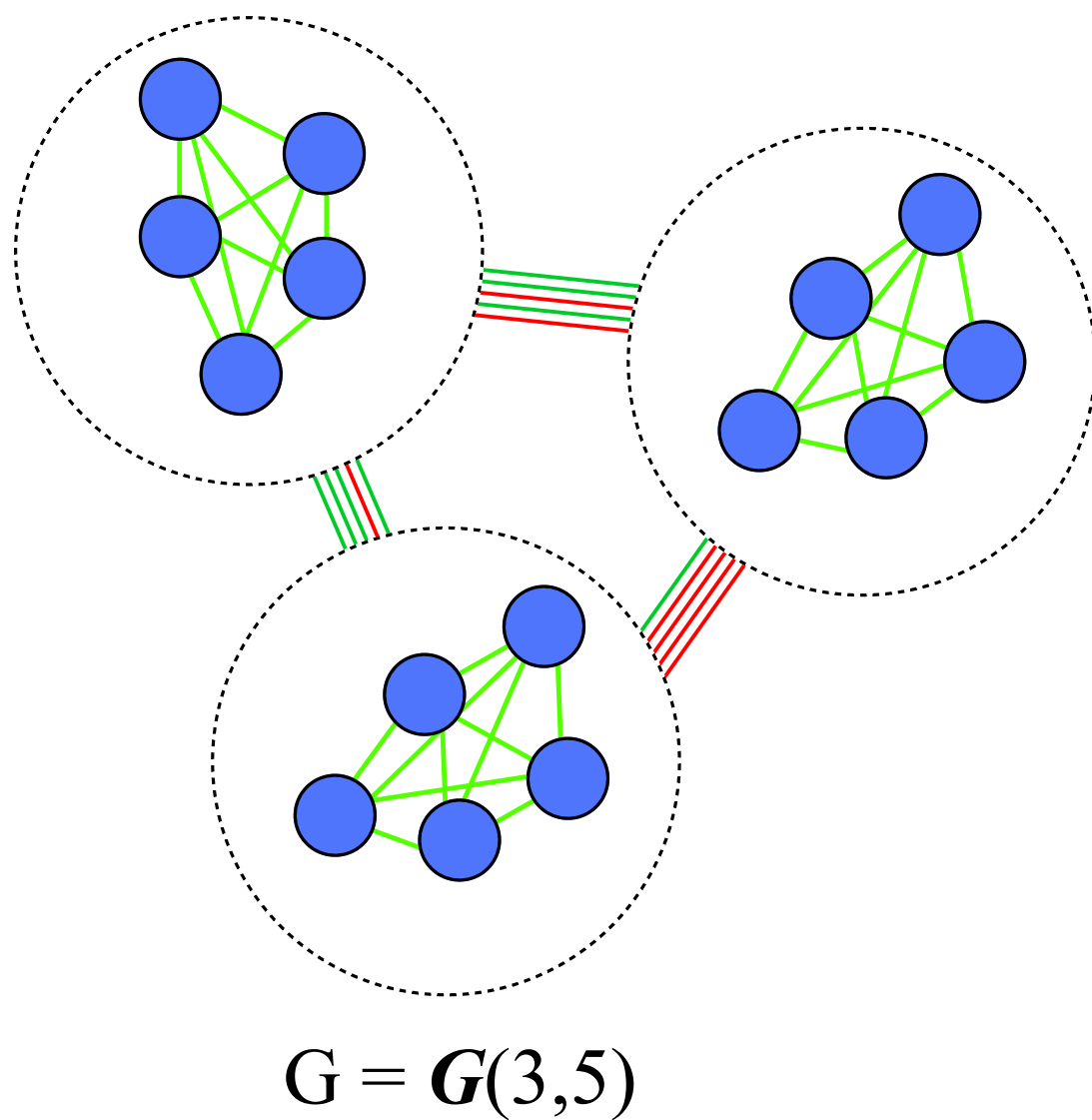
Abstract

We study opinion dynamics on networks with a nontrivial community structure, assuming individuals can update their binary opinion as the result of the interactions with an external influence with strength $h \in [0, 1]$ and with other individuals in the network. To model such dynamics, we consider the Ising model with an external magnetic field on a family of finite networks with a clustered structure. Assuming a unit strength for the interactions inside each community, we assume that the strength of interaction across different communities is described by a scalar $\epsilon \in [-1, 1]$, which allows a weaker but possibly antagonistic effect between communities. We are interested in the stochastic evolution of this system described by a Glauber-type dynamics parameterized by the inverse temperature β . We focus on the low-temperature regime $\beta \rightarrow \infty$, in which homogeneous opinion patterns prevail and, as such, it takes the network a long time to fully change opinion. We investigate the different metastable and stable states of this opinion dynamics model and how they depend on the values of the parameters ϵ and h . More precisely, using tools from statistical physics, we derive rigorous estimates in probability, expectation, and law for the first hitting time between metastable (or stable) states and (other) stable states, together with tight bounds on the mixing time and spectral gap of the Markov chain describing the network dynamics. Lastly, we provide a full characterization of the critical configurations for the dynamics, i.e., those which are visited with high probability along the transitions of interest.

- Created **graphs** to represent social networks
- Defined **non-hostile** and **adversarial** interactions mathematically
- Used a popular statistical physics model (the metropolis algorithm) on the graphs
- Defined “meta-stable” states, and identified signifiers of critical transitions

This paper rigorously utilises graph theory and the ising model

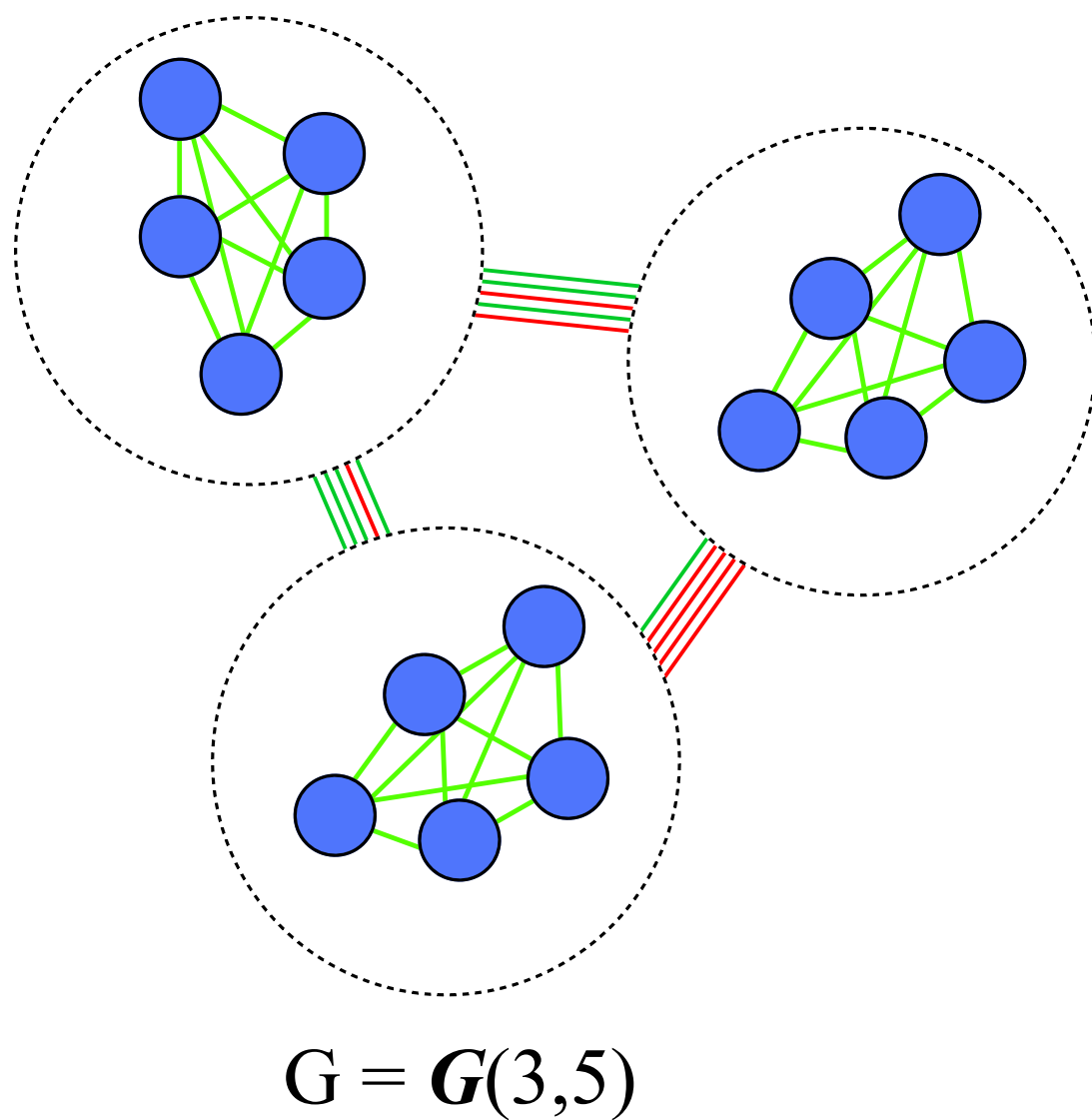
Modelling social networks as graphs



- $G(n,k)$: Graph with ‘ k ’ complete **subgraphs** (clusters) of size ‘ n ’
- Each node in subgraph is connected to its ‘image’ in every other subgraph
- Connections inside a subgraph are all **positive**
- Inter-cluster connections can be both **positive** or **negative**
- Number of connections each node has is ‘ $n-k-2$ ’ (6 in this case)

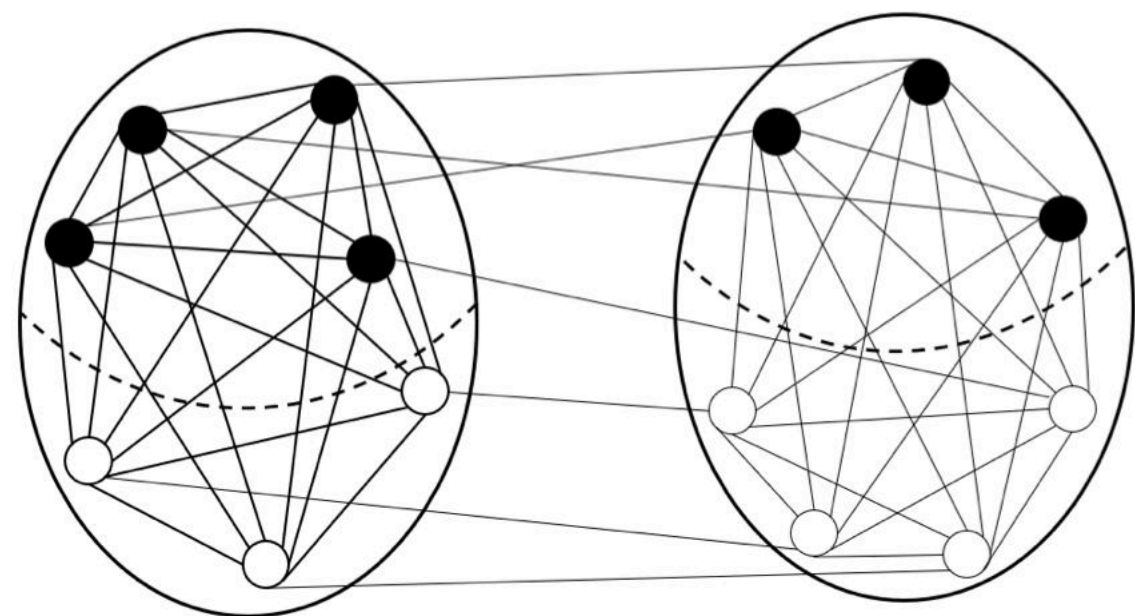
More specifically, for every $k \geq 2$ and every $n \geq 2$ we consider an undirected graph $G = \mathcal{G}(k, n)$ consisting of k clusters, each of which is a complete subgraph of size n , in which we further connect each node, $i = 1, \dots, n$ also to its $k - 1$ “twins” in the other $k - 1$ clusters (those whose labels have the same remainder modulo n), hence obtaining a regular graph where each node has degree $n + k - 2$.

Modelling social networks as graphs



- Each cluster is a community, with the nodes being individual users.
- Inside a community, opinions are agreed upon and the edges are thus green
- Inter-community beliefs can be contested, and thus some edges are red.
- Belief compatibility is parametrized by ' $\epsilon \in [-1,1]$ ' where $\epsilon = 1$ for compatibility and $\epsilon = (-1)$ for incompatibility
- $\epsilon = 1$ inside a cluster

Modelling social networks as graphs



$$G = G(2,7)$$

$$H(\sigma) := - \sum_{(i,j) \in E_{\text{int}}} \sigma_i \sigma_j - \epsilon \sum_{(i,j) \in E_{\text{cross}}} \sigma_i \sigma_j - h \sum_{i \in V} \sigma_i,$$

- Now, each node is assigned a spin $\sigma(i) \in \{-1, +1\}$ randomly (these can be interpreted as original individual beliefs)
- Now, an ‘Interaction Hamiltonian (H)’ is defined, interpreted as the ‘energy’
- Energy of internal clusters is just the sum of spins
- Energy for cross clusters have a factor of scalar ϵ
- External influence is mapped to $h \in [0,1]$ (interpreted as political climate)

The Metropolis algorithm

```
For node in graph :  
    flip spin of node  
    if energy of graph lowers:  
        flip spin permanently with probability =  $p(T, \Delta E)$   
    else:  
        undo the flip
```

With this setup, Baldassari et al. have shown a lot of interesting results.

Results of the Baldassari experiment

- Meta-stable states when external temperature goes to zero $\rightarrow$ low opinion changes when social climate is not intense
- Also mapped co-operation between communities. When $\epsilon > 0$, both communities have similar opinions (two metastable states; both negative or positive) and when $\epsilon < 0$, four metastable states ($[+,+]$, $[+,-]$, $[-,-]$, $[-,+]$)
- **Mathematically rigorously find the ‘states’ the system hits before undergoing critical transitions**

Results of the Baldassari experiment

- Formalised structure of external influences on the overall state of the social network
- Proved that antagonism can live for exponentially longer lifetimes if $\epsilon > 0$, once opposing viewpoints reach a metastable state
- The cluster structure plays a huge role in the amount of meta-stable states that a network can have.